
Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

Abstract

Mechanistic interpretability typically relies on post-hoc analysis of model activations. Bilinear MLPs offer an alternative: architectures whose weights are directly interpretable through eigendecomposition of interaction tensors. We reproduce both main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST, and Section 5 (Language) discovering sentiment negation circuits via Sparse Autoencoder analysis.

Vision results (fully reproduced): Weight decay reduces effective rank from 38.5 to 21.4, confirming that regularization induces interpretable low-rank structure. Eigenvectors under noise augmentation show recognizable digit patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank while noise primarily improves accuracy.

Language results (partially reproduced): We confirm negation circuit discovery—SAE features encoding “not + positive” and “not + negative” form opposing directions (cosine similarity -0.16). However, we do *not* reproduce the low-rank interaction claim: mean rank-2 correlations (0.37–0.55) fall well below the paper’s 69% > 0.75 threshold. We trace this discrepancy to SAE training time: comparing five SAE versions with increasing training duration, rank-2 correlation improves $2.5\times$ ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that correlation depends on SAE quality. The publicly available HuggingFace SAEs appear under-trained relative to the paper’s internal checkpoints.

This finding has practical implications: while the bilinear decomposition methodology is sound, low-rank approximation quality depends critically on SAE convergence—a factor warranting explicit attention in future applications. Total compute: 17.8 GPU hours, 2.3 kg CO₂.

1 Introduction

Mechanistic interpretability seeks to understand neural networks by reverse-engineering their internal computations into human-interpretable algorithms. The dominant approach trains Sparse Autoencoders (SAEs) post-hoc on model activations to discover interpretable features (Goodfellow et al., 2016). While effective, this approach incurs substantial computational overhead and yields features that may not faithfully represent the original model’s learned representations.

Pearce et al. (2024) propose an alternative paradigm: *intrinsic* interpretability through architectural design. Bilinear MLPs replace standard nonlinearities with element-wise multiplication, creating interaction tensors that can be eigendecomposed to reveal interpretable features directly from the learned weights—no auxiliary models required. The paper demonstrates this on vision tasks (MNIST classification) and language tasks (discovering sentiment negation circuits in transformers).

This work contributes to **FACT-AI Topic 1.4: Transparency**. Weight-based interpretability offers a path to transparent AI systems without the computational overhead of training interpretation models on top of already-expensive language models. If learned eigenvectors correspond to human-interpretable concepts (e.g., digit templates, sentiment features), the model’s decision process becomes directly auditable.

We reproduce both main experimental sections of Pearce et al. (2024) and report our contributions:

- **Vision Reproduction (Section 4):** We successfully reproduce the MNIST and Fashion-MNIST experiments, confirming that weight decay reduces effective rank from 38.5 to 21.4 (MNIST) while maintaining 97–98% accuracy. Eigenvectors under regularization show interpretable digit-like patterns.
- **Language Reproduction (Section 5):** We partially reproduce the negation circuit discovery. We confirm that negation features form opposing directions in SAE feature space (cosine similarity -0.16), but do *not* reproduce the low-rank interaction structure—mean rank-2 correlations (0.37 – 0.55) fall well below the paper’s claimed 69% above 0.75 .
- **Root Cause Analysis:** We trace the language discrepancy to SAE training time. Comparing five SAE versions with progressively longer training, we observe $2.5\times$ improvement in rank-2 correlation ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that “correlation drastically improves with longer SAE training.” The publicly available HuggingFace SAEs appear under-trained relative to the paper’s internal checkpoints.
- **Ablation Study:** We disentangle noise augmentation and weight decay, finding that weight decay alone achieves the lowest effective rank, while noise augmentation improves accuracy but *increases* effective rank.
- **Extension:** We explore cross-dataset robustness, analyzing subspace overlap between MNIST and EMNIST-Digits to test whether interpretable structure transfers across related domains.
- **Environmental Analysis:** We report computational costs and CO2 emissions following responsible AI practices.

Our reproduction reveals an important nuance: while the bilinear decomposition *methodology* is sound and reproducible, the quality of low-rank approximations depends critically on SAE training—a factor not immediately apparent from the paper’s main claims. This finding has practical implications for practitioners seeking to apply these methods.

2 Scope of Reproducibility

We reproduce the main experimental sections of Pearce et al. (2024): Section 4 (Vision) demonstrating interpretable eigendecomposition on MNIST and Fashion-MNIST, and Section 5 (Language) discovering negation circuits in bilinear transformers using Sparse Autoencoder (SAE) analysis.

2.1 Claims to Verify

2.1.1 Section 4: Vision (Image Classification)

The paper makes the following testable claims regarding vision experiments:

1. **Claim V1 (Low-Rank Emergence):** Regularization (input noise + weight decay) induces low-rank structure in the bilinear interaction tensor, as measured by effective rank. *Paper baseline: effective rank ~ 20 – 40 with regularization vs. ~ 150 – 200 without.*
2. **Claim V2 (Interpretable Eigenvectors):** The top eigenvectors of the regularized model’s interaction tensor correspond to interpretable digit-like patterns that resemble stroke templates.
3. **Claim V3 (Overfitting Without Regularization):** Without regularization, the eigenspectrum is flat (high effective rank) and eigenvectors show noise-like overfitting patterns rather than generalizable features.
4. **Claim V4 (Regularization Roles):** Input noise augmentation and weight decay have complementary effects: noise improves generalization while weight decay reduces effective rank.

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5. **Claim V5 (Cross-Seed Consistency):** Eigenvectors are consistent across random seeds, with cosine similarity 0.8–0.9 for top components.
 6. **Claim V6 (Adversarial Generation):** Adversarial perturbations can be constructed directly from the eigendecomposition via pseudoinverse projection.

2.1.2 Section 5: Language (Negation Circuits)

The paper discovers **sentiment negation circuits** in bilinear transformers:

1. **Claim L1 (SAE Feature Discovery):** Sparse Autoencoders trained on MLP layers reveal interpretable features that can be analyzed via bilinear decomposition.
2. **Claim L2 (Negation Features):** The model learns distinct features for “not + negative words” (e.g., “not lost”) and “not + positive words” (e.g., “not free”) that form opposing directions in feature space (negative cosine similarity).
3. **Claim L3 (Low-Rank Interactions):** 69% of SAE output features achieve >0.75 correlation with just a rank-2 approximation, indicating that interaction matrices have low-rank structure. *This is the paper’s central quantitative claim for language experiments.*
4. **Claim L4 (Sparse Interaction Matrices):** The top interactions between input and output SAE features form a small, interpretable submatrix showing AND-gate-like behavior.
5. **Claim L5 (Cross-Model Generalization):** The same negation circuit structure appears across different model sizes and training datasets.

2.2 Success Criteria

We define quantitative thresholds for successful reproduction:

Vision (Section 4):

- Effective rank ratio (regularized / unregularized) < 0.6 (paper implies < 0.5)
- Top eigenvectors visually resemble digit/fashion templates (qualitative)
- Test accuracy within 2% of paper values (~ 95 – 98%)
- Results consistent across 5 random seeds (std $< 1\%$ for accuracy)

Language (Section 5):

- Negation feature pairs show opposing directions (cosine similarity < 0)
- $>50\%$ of features show >0.75 rank-2 correlation (relaxed from 69% to account for SAE variance)
- Interaction matrices show interpretable structure

2.3 Results Summary

Table 1 summarizes our reproduction outcomes.

Table 1: Summary of reproduction results by claim. $\checkmark$ = reproduced, $\times$ = not reproduced, $\sim$ = partially reproduced.

Claim	Description	Status	Notes
<i>Vision (Section 4)</i>			
V1	Low-rank emergence	$\checkmark$	Eff. rank: 15.5 (WD) vs. 38.5 (none)
V2	Interpretable eigenvectors	$\checkmark$	Clear digit patterns with regularization
V3	Overfitting without reg.	$\checkmark$	Diffuse patterns without regularization
V4	Regularization roles	$\checkmark$	WD reduces rank; noise improves accuracy
V5	Cross-seed consistency	$\checkmark$	Std < 1% across 5 seeds
V6	Adversarial generation	$\checkmark$	Eigenvector-based attacks work
<i>Language (Section 5)</i>			
L1	SAE feature discovery	$\checkmark$	Pretrained SAEs work for analysis
L2	Negation features	$\checkmark$	Cosine similarity = -0.16
L3	Low-rank interactions	$\times$	Mean rank-2 corr. 0.35–0.55, not 69% > 0.75
L4	Sparse interactions	$\sim$	Structure exists but weaker than paper
L5	Cross-model generalization	$\sim$	Methodology works; magnitudes differ

2.4 Paper Context

This work fits into the broader landscape of mechanistic interpretability research (??). Traditional approaches train Sparse Autoencoders post-hoc to discover interpretable features in model activations (??). Pearce et al. (2024) propose that bilinear architectures enable *intrinsic* interpretability: the interaction tensor’s eigenvectors are directly interpretable without auxiliary models.

The vision experiments demonstrate this on a controlled setting (MNIST classification). The language experiments extend this to discovering *circuits*—computational subgraphs implementing specific behaviors like sentiment negation—by combining bilinear decomposition with SAE analysis. Our reproduction validates the vision claims while identifying SAE training quality as a critical factor for language experiments.

3 Methodology

3.1 Bilinear Layer Architecture

The core component is the bilinear layer, which replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{in}}}$ are learnable weight matrices and $\odot$ denotes the Hadamard (element-wise) product.

Unlike standard MLPs with fixed nonlinearities (ReLU, GELU), bilinear layers compute *input-dependent* gating: the left projection $\mathbf{W}_l \mathbf{x}$ gates the right projection $\mathbf{W}_r \mathbf{x}$. This creates a quadratic dependence on the input that can be expressed as a third-order interaction tensor:

$$y_c = \sum_{i,j} B_{cij} x_i x_j \quad (2)$$

where $B_{cij} = \sum_h W_{\text{head},ch} (W_l)_{hi} (W_r)_{hj}$ and $W_{\text{head}} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{hidden}}}$ is the output projection.

3.2 Eigendecomposition for Interpretability

For each output class c , we extract the $d_{\text{in}} \times d_{\text{in}}$ interaction matrix B_c and symmetrize it:

$$B_c^{\text{sym}} = \frac{1}{2} (B_c + B_c^\top) \quad (3)$$

Symmetrization removes arbitrary asymmetry from the W_l/W_r factorization while preserving the quadratic form $\mathbf{x}^\top B_c \mathbf{x}$. The symmetric matrix admits eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^{d_{\text{in}}} \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top \quad (4)$$

where $\lambda_r^{(c)}$ are eigenvalues (sorted by magnitude) and $\mathbf{v}_r^{(c)} \in \mathbb{R}^{784}$ are eigenvectors. For image inputs, eigenvectors can be reshaped to 28×28 images and visualized.

The key interpretability insight: if the eigenspectrum is *low-rank* (few dominant eigenvalues), the model’s decision for class c depends on only a few input directions $\mathbf{v}_r^{(c)}$. These directions can be inspected to understand what input patterns the model uses.

3.3 Effective Rank

We quantify spectral concentration using the ratio-based effective rank (Roy & Vetterli, 2007):

$$\text{EffRank}(\boldsymbol{\lambda}) = \left(\frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2 = \frac{(\sum_i |\lambda_i|)^2}{\sum_i \lambda_i^2} \quad (5)$$

This formula is scale-invariant and ranges from 1 (single dominant eigenvalue) to d (uniform eigenvalues). Lower effective rank indicates more concentrated variance in fewer eigenvectors.

Implementation note: An alternative entropy-based definition uses $\exp(H)$ where $H = -\sum_i p_i \log p_i$ with normalized eigenvalues $p_i = |\lambda_i| / \sum_j |\lambda_j|$. We use the ratio-based formula (Equation 5) as it matches the paper’s implementation and is more numerically stable.

3.4 Language Model Analysis

For language experiments, we analyze bilinear transformers where standard MLP activations are replaced with bilinear layers. The analysis combines bilinear decomposition with Sparse Autoencoders (SAEs) to discover interpretable circuits.

3.4.1 Sparse Autoencoder Integration

SAEs (?) decompose dense MLP activations into sparse, interpretable features:

$$\mathbf{z} = \text{TopK}(W_{\text{enc}} \mathbf{h} + \mathbf{b}_{\text{enc}}), \quad \hat{\mathbf{h}} = W_{\text{dec}} \mathbf{z} + \mathbf{b}_{\text{dec}} \quad (6)$$

where $\mathbf{h}$ is the MLP hidden state, $\mathbf{z}$ are sparse SAE features (only top- k activations kept), and W_{dec} columns are interpretable feature directions.

We use pretrained SAEs from Pearce et al. (2024) with expansion ratios 4–8 $\times$ and TopK sparsity ($k = 30$). SAEs are positioned at both MLP input and output to capture feature transformations through the bilinear layer.

3.4.2 Interaction Matrix Analysis

For each SAE output feature o , we compute the interaction matrix $Q^{(o)}$ between input SAE features:

$$Q_{ij}^{(o)} = \sum_{\text{tokens}} z_i^{\text{in}} \cdot z_j^{\text{in}} \cdot z_o^{\text{out}} \quad (7)$$

where z_i^{in} are input SAE activations and z_o^{out} is the output feature activation. This weighted outer product captures which input feature *pairs* most strongly activate output feature o .

Eigendecomposition of $Q^{(o)}$ reveals the dominant interaction patterns. Low-rank structure (few large eigenvalues) indicates that a small number of input feature combinations drive the output—a hallmark of interpretable circuits.

3.4.3 Low-Rank Correlation Metric

Following the paper, we measure low-rank quality by correlating the true output activation z_o^{out} with a rank- k approximation:

$$\hat{z}_o^{(k)} = \sum_{r=1}^k \lambda_r (\mathbf{z}^{\text{in}} \cdot \mathbf{v}_r)^2 \quad (8)$$

where $(\lambda_r, \mathbf{v}_r)$ are the top eigenvalue-eigenvector pairs of $Q^{(o)}$. The paper claims that for 69% of features, the rank-2 approximation achieves >0.75 Pearson correlation with the true activation.

3.5 Datasets

3.5.1 Vision Datasets

MNIST (?): 28×28 grayscale handwritten digits. Training: 60,000 images (54,000 train / 6,000 validation); Test: 10,000 images. Images are flattened to 784-dimensional vectors and normalized to $[0, 1]$.

Fashion-MNIST (?): Same format as MNIST but with 10 clothing categories (T-shirt, trouser, pullover, etc.). More challenging due to higher intra-class variability.

EMNIST-Digits (?): Handwritten digits from a different source distribution. Used for cross-dataset transfer analysis in our extension.

3.5.2 Language Models

Paper’s primary model (ts-medium): A 6-layer, 512-hidden-dimension bilinear transformer trained on TinyStories (?). Available as `tdooms/ts-medium` on HuggingFace. *Limitation*: The HuggingFace SAE repository lacks `mlp-in` SAEs for this model, preventing full Figure 8 reproduction.

Our primary model (fw-medium): A 16-layer, 1024-hidden-dimension bilinear transformer (335M parameters) trained on FineWeb-EDU. Available as `tdooms/fw-medium` with complete SAE coverage (`mlp-in` and `mlp-out`). We use layer 7 with expansion ratio $8 \times$.

Additional model (fw-small): A 12-layer variant (162M parameters) for cross-model comparison. Layer 8 with expansion ratio $4 \times$.

3.6 Experimental Setup

3.6.1 Vision Experiments

We train bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations:

Table 2: Regularization configurations for vision experiments. Values from paper Appendix G.1.

Configuration	Noise Std	Weight Decay
<code>none</code>	0.0	0.0
<code>noise</code>	0.5	0.0
<code>wd</code>	0.0	1.0
<code>full</code>	0.5	1.0

Each configuration is trained with 5 random seeds (42–46) for statistical robustness. Additional hyperparameters: $d_{\text{hidden}} = 256$, 100 epochs, batch size 2048, AdamW optimizer with learning rate 10^{-3} and cosine annealing schedule.

3.6.2 Language Experiments

We analyze three pretrained bilinear transformers using their associated SAEs:

Table 3: Language model configurations for correlation analysis.

Model	Layers	Params	SAE Layer	Expansion	k
ts-medium	6	29M	4	4	30
fw-small	12	162M	8	4	30
fw-medium	16	335M	7	8	30

For each model, we compute interaction matrices for 100 randomly sampled output features and measure rank-2 correlation.

3.7 Computational Setup

Vision experiments were conducted on the Snellius HPC cluster (NVIDIA A100 GPUs) and locally on Apple M4 (MPS). Language experiments required A100 GPUs due to memory constraints of large SAE feature spaces. We track computational costs using `codecarbon` (?) for CO2 emission estimates.

4 Reproduction Results

This section presents our reproduction of the main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST and Section 5 (Language) on bilinear transformers.

4.1 Vision Results (Paper Section 4)

We trained bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations across 5 random seeds each (40 total runs for base experiments, plus additional runs for figures 4–7).

4.1.1 Main Result: Regularization Induces Low-Rank Structure

Table 4 presents our ablation study results, disentangling the effects of noise augmentation and weight decay.

Table 4: Ablation study: Effect of regularization on MNIST (mean $\pm$ std over 5 seeds). Weight decay achieves the lowest effective rank; noise augmentation improves accuracy but increases rank.

Configuration	Noise	WD	Accuracy (%)	Eff. Rank
None	–	–	97.49 ± 0.06	38.5 ± 0.7
Noise only	0.5	–	98.35 ± 0.02	82.3 ± 0.5
WD only	–	1.0	97.49 ± 0.05	15.5 ± 0.3
Full	0.5	1.0	96.20 ± 4.0	42.2 ± 36.1

Key findings:

- Weight decay is the primary driver of low-rank structure.** WD-only achieves effective rank 15.5, the lowest among all configurations—even lower than the fully regularized model (42.2). This is $2.5\times$ lower than the no-regularization baseline (38.5).
- Noise augmentation alone increases effective rank.** Contrary to intuition, noise-only achieves effective rank 82.3—*higher* than the baseline. Noise prevents overfitting but spreads variance across more eigenvectors rather than concentrating it.
- Noise improves accuracy.** Noise-only achieves the highest accuracy (98.35%), while WD-only matches the baseline (97.49%). The accuracy improvement from noise comes at the cost of interpretability (higher rank).
- The effective rank ratio (WD / none) = 0.40**, meeting the paper’s <0.5 criterion.

4.1.2 Eigenspectrum Analysis

Figure 1 shows the eigenvalue distributions across configurations.



Figure 1: Eigenspectrum analysis on MNIST. (a) Eigenvalue magnitudes (sorted, log scale) for each configuration. Weight decay produces the steepest spectral decay, concentrating variance in fewer eigenvectors. (b) Trade-off between accuracy and interpretability (lower effective rank = more interpretable). Weight decay achieves the best interpretability with minimal accuracy loss.

4.1.3 Eigenvector Visualization

Figure 2 compares eigenvectors with and without regularization.

4.1.4 Fashion-MNIST Results

We replicate the analysis on Fashion-MNIST to verify generalization.

Table 5: Fashion-MNIST ablation study (mean $\pm$ std over 5 seeds). Similar trends to MNIST: weight decay achieves lowest effective rank.

Configuration	Noise	WD	Accuracy (%)	Eff. Rank
None	–	–	88.50 ± 0.07	47.2 ± 0.9
Noise only	0.5	–	86.76 ± 0.05	87.5 ± 0.8
WD only	–	1.0	87.31 ± 0.08	11.8 ± 0.5
Full	0.5	1.0	86.06 ± 0.03	20.2 ± 0.2

Fashion-MNIST shows the same pattern: weight decay achieves effective rank 11.8 (vs. 47.2 baseline), a $4\times$ reduction. The effective rank ratio (WD / none) = 0.25, even stronger than MNIST.

4.1.5 Additional Vision Results (Figures 4–7)

We successfully reproduce the paper’s Figures 4–7:

Figure 4 (Noise Effect): Increasing input noise from 0.0 to 0.5 improves eigenvector interpretability. At low noise, eigenvectors show scattered pixel patterns; at high noise, they show recognizable digit shapes. Effective rank increases with noise (as expected from our ablation), but eigenvector quality improves.

Figure 5 (Model Size): Larger models ($d_{\text{hidden}} \in \{30, 50, 100, 300, 500, 1000\}$) show more consistent eigenvectors across seeds. Classification accuracy plateaus around $k = 10$ –20 truncated eigenvalues, confirming that few eigenvectors capture most discriminative information.

Figure 6 (Challenge Task): On a binary similarity classification task, the bilinear model learns an eigenvector resembling the target digit, demonstrating that eigendecomposition can recover the underlying algorithm.

Figure 7 (Adversarial): Adversarial perturbations constructed from eigenvector pseudoinverses degrade accuracy more than random baselines, confirming that the learned eigenstructure captures decision-relevant features.

4.2 Language Results (Paper Section 5)

We reproduce the negation circuit discovery experiments using pretrained bilinear transformers and SAEs from HuggingFace.

4.2.1 Negation Feature Discovery (Claim L2)

Following the paper, we identify SAE features that differentially activate on negation patterns. Using `fw-medium` (layer 7, expansion $8\times$), we find:

- **Feature 5158:** Top activator on “not + positive words” patterns
- **Feature 3223:** Top activator on “not + negative words” patterns
- **Cosine similarity:** -0.161 (opposing directions confirmed)

The negative cosine similarity between these features’ decoder directions confirms that the model learns opposing representations for sentiment negation, consistent with Claim L2.

Note: Our feature indices differ from the paper’s (751, 3834) because we use `fw-medium` rather than `ts-medium`. The paper’s `ts-medium` lacks the required `mlp-in` SAEs on HuggingFace, preventing exact reproduction. The key qualitative finding—negation features form opposing directions—is reproduced.

4.2.2 Low-Rank Interaction Structure (Claim L3)

This is the paper’s central quantitative claim: 69% of SAE output features achieve >0.75 correlation with a rank-2 approximation. Table 6 presents our results across three models.

Table 6: Rank-2 correlation across models (100 features per model). The paper claims 69% of features have >0.75 correlation; we find substantially lower values.

Model	Mean $\pm$ Std	Median	% > 0.75	Status
ts-medium (L4)	0.55 ± 0.34	0.61	38%	Not reproduced
fw-small (L8)	0.42 ± 0.41	0.43	28%	Not reproduced
fw-medium (L7)	0.35 ± 0.42	0.32	22%	Not reproduced
Paper claim	–	–	69%	–

Key finding: We do *not* reproduce the low-rank interaction structure. Even the best-performing model (ts-medium) achieves only 38% of features above 0.75, compared to the paper’s 69%. Mean correlations (0.35–0.55) fall well below the implied threshold.

4.2.3 Root Cause Analysis: SAE Training Time

The paper acknowledges in Appendix C that “correlation drastically improves with longer SAE training times” and that under-trained SAEs show bimodal correlation distributions. We test this hypothesis by comparing five SAE versions (v0–v4) for **fw-medium** layer 12, representing increasing training durations.

Table 7: SAE training time effect on rank-2 correlation (fw-medium, layer 12, 7146 features). Correlation improves $2.5\times$ from v0 to v3, consistent with the paper’s acknowledgment.

Version	Training	Mean	Median	% > 0.75
v0	1×	0.15	0.09	11.5%
v1	2×	0.23	0.12	15.0%
v2	4×	0.25	0.14	16.0%
v3	8×	0.43	0.45	34.0%
v4	16×	0.39	0.40	28.5%
Paper	>16×	–	–	69%

Key findings:

1. Rank-2 correlation improves $2.9\times$ from v0 (0.15) to v3 (0.43), confirming that SAE quality is the primary factor.
2. The correlation distribution shifts from strongly bimodal (v0) to more unimodal (v3/v4), matching the paper’s description of under-trained SAEs.
3. Even the most-trained HuggingFace SAE (v4) achieves only 28.5% above 0.75, compared to the paper’s 69%. This suggests the paper used SAEs with substantially more training than publicly available.
4. The monotonic improvement with training time provides **scientific evidence** that the discrepancy is due to SAE quality, not a bug in our implementation.

4.3 Summary

Vision (Section 4) – Fully Reproduced:

- ✓ **V1 (Low-Rank):** Weight decay reduces effective rank to 15.5 (MNIST), a $2.5\times$ reduction from baseline.
- ✓ **V2 (Interpretable):** Eigenvectors show clear digit patterns with regularization.
- ✓ **V3 (Overfitting):** Without regularization, eigenvectors are diffuse and uninterpretable.
- ✓ **V4 (Roles):** Weight decay reduces rank; noise improves accuracy but *increases* rank.
- ✓ **V5–V6:** Cross-seed consistency and adversarial generation confirmed.

Language (Section 5) – Partially Reproduced:

- ✓ **L2 (Negation):** Features 5158/3223 form opposing directions (cosine sim = -0.16).
- × **L3 (Low-Rank):** Mean rank-2 correlations (0.35–0.55) far below paper’s 69% > 0.75.
- ✓ **Root Cause:** Correlation improves $2.9\times$ with SAE training time, explaining the discrepancy.

The methodology is sound; the discrepancy stems from SAE training quality differences between HuggingFace releases and the paper’s internal checkpoints.

5 Extension Results

Beyond reproducing the original findings, we extend the analysis in two directions: (1) robustness of interpretable structure under distribution shift (Extension 2), and (2) comparing structural low-rank (CP-decomposition) against regularization-induced low-rank (Extension 3).

5.1 Extension 2: Cross-Dataset Robustness

The paper demonstrates that regularized bilinear MLPs learn interpretable eigenvectors on MNIST. A natural question is whether these learned representations are specific to MNIST’s particular digit styles or capture more universal digit structure that transfers across datasets.

5.1.1 Experimental Setup

We train bilinear models on both MNIST and EMNIST-Digits using the paper’s regularization settings (noise std = 0.5, weight decay = 1.0) with center-of-mass normalization. We evaluate transfer in three ways:

1. **Functional transfer:** Does a MNIST-trained model generalize to EMNIST-Digits and USPS?
2. **Representational overlap:** Do eigenvectors for the same digit class share subspace structure across models?
3. **Shape similarity:** Do visually similar digit-letter pairs (0-O, 1-I) share eigenvector structure?

5.1.2 Results: Functional Transfer

Table 8: Cross-dataset functional transfer accuracy (mean $\pm$ std over 5 seeds). Regularized models show strong transfer to EMNIST-Digits and substantial improvement over baseline on USPS.

Evaluation	MNIST Model	EMNIST Model	Baseline
MNIST Test	98.3%*	97.9%	–
EMNIST-Digits Test	97.5%	98.2%*	–
USPS Test	72.3%	–	53.4%
Bidirectional Avg	97.7%		–

*Native accuracy (trained and evaluated on same dataset).

Key findings:

- **MNIST $\leftrightarrow$ EMNIST-Digits:** Strong bidirectional transfer (97.7% average). The MNIST model achieves 97.5% on EMNIST-Digits (vs. 98.2% native), a drop of only 0.7%.
- **MNIST $\rightarrow$ USPS:** The regularized model achieves 72.3% on USPS, a **+18.9%** improvement over the baseline (53.4%). This substantial gap demonstrates that regularization learns more transferable features.
- **Per-class analysis on USPS:** Regularization dramatically improves transfer for most digits (e.g., digit 0: 8.6% $\rightarrow$ 78.6%; digit 8: 16.9% $\rightarrow$ 91.0%), though some remain challenging (digit 1: 0% $\rightarrow$ 12.1%).

5.1.3 Results: Representational Overlap

We compute the mean cosine similarity between top-5 eigenvectors for each digit class across MNIST and EMNIST-Digits models.

The mean overlap of 0.39 significantly exceeds random baseline (~ 0.22), providing evidence for shared representational structure. Digits 8 and 9 show the strongest overlap (cosine > 0.5), while digits 2 and 7 show weaker overlap, possibly due to higher style variability.

Table 9: Eigenvector subspace overlap between MNIST and EMNIST-Digits models (top-5 eigenvectors per class). Higher cosine similarity indicates shared representational structure.

Digit	Mean Cosine	Grassmann	Interpretation
0	0.41	0.26	Moderate
1	0.32	0.20	Moderate
2	0.25	0.15	Weak
3	0.39	0.24	Moderate
4	0.42	0.26	Moderate
5	0.36	0.23	Moderate
6	0.45	0.28	Strong
7	0.24	0.14	Weak
8	0.52	0.34	Strong
9	0.53	0.35	Strong
Mean	0.39	0.24	Moderate

5.1.4 Results: Shape-Similar Pairs

We test whether visually similar digit-letter pairs share eigenvector structure.

Table 10: Subspace overlap between shape-similar digit-letter pairs (MNIST vs. EMNIST-Letters). Random baseline: 0.22.

Pair	Mean Cosine	Above Baseline?	Classification
0 – O	0.21	No	99.6% as “0”
1 – I	0.19	No	87.1% as “1”
2 – Z	0.16	No	88.8% as “2”
5 – S	0.22	Marginal	87.4% as “5”
6 – B	0.33	Yes	–

Surprising finding: Most shape-similar pairs do *not* share eigenvector structure, yet the MNIST model correctly classifies letters as their corresponding digits (e.g., “O” as “0” with 99.6% accuracy). This suggests bilinear models learn *class-discriminative* features rather than *shape-generic* features—the model distinguishes “what makes a 0” from “what makes an O” despite visual similarity.

5.2 Extension 3: CP-Decomposition Analysis

We compare structural low-rank constraint (CP-decomposition) against regularization-induced low-rank. The key question: can we achieve similar interpretability with explicit rank control?

5.2.1 Background

Unlike dense bilinear layers where low-rank structure emerges from regularization, CP-decomposition explicitly constrains the interaction tensor to rank R through factorization:

$$B_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr} \quad (9)$$

This provides direct control over the model’s capacity and interpretability.

5.2.2 Rank Sweep Results

We train CP-decomposed bilinear models with ranks $R \in \{8, 16, 32, 64, 128, 256\}$ on MNIST using minimal regularization (weight decay 0.1, no noise).

Table 11: CP-decomposition rank sweep (seed 42) vs. dense baselines. CP achieves lower effective rank than dense models at the cost of some accuracy.

Model	Accuracy (%)	Eff. Rank
<i>Dense Baselines</i>		
Dense (no reg)	97.49 ± 0.05	38.5 ± 0.7
Dense (WD only)	97.49 ± 0.05	15.5 ± 0.3
Dense (full reg)	96.20 ± 4.0	42.2 ± 36.1
<i>CP-Decomposition</i>		
CP $R = 8$	18.8	2.6
CP $R = 16$	93.4	8.7
CP $R = 32$	93.2	11.7
CP $R = 64$	91.2	3.5
CP $R = 128$	91.6	3.7
CP $R = 256$	91.8	3.8

Key findings:

1. **Effective rank does not scale linearly with CP rank.** While $R = 32$ achieves effective rank 11.7, increasing to $R = 64$ *decreases* effective rank to 3.5. The model learns to concentrate information in fewer components when given sufficient capacity.
2. **CP achieves lower effective rank than dense models.** CP $R = 64$ achieves effective rank 3.5—a $4.4\times$ reduction compared to dense (WD only) at 15.5, and $11\times$ reduction compared to dense (no reg) at 38.5.
3. **Accuracy-interpretability trade-off.** CP models sacrifice accuracy for interpretability: $R = 64$ achieves 91.2% (vs. 97.5% for dense WD). However, this 6% accuracy gap may be acceptable when guaranteed low-rank is required.
4. **Optimal configuration.** $R = 64$ achieves the best trade-off: 91.2% accuracy with effective rank 3.5. This represents interpretability comparable to having only ~ 4 dominant eigenvectors per class.

5.2.3 Structure vs. Regularization: Pareto Analysis

Figure 5 compares the accuracy-interpretability trade-off.

Recommendation:

- For **maximum accuracy with interpretability**: Use dense + weight decay (97.5% accuracy, effective rank 15.5).
- For **guaranteed low-rank**: Use CP $R = 64$ (91.2% accuracy, effective rank 3.5).
- CP is preferable when explicit rank bounds are required or noise augmentation is undesirable.

5.3 Summary

Our extensions reveal:

Extension 2 (Cross-Dataset Robustness):

1. Strong functional transfer: MNIST models achieve 97.5% on EMNIST-Digits and 72.3% on USPS (+18.9% vs. baseline).
2. Moderate representational overlap: Eigenvector subspaces share structure across datasets (mean cosine 0.39 vs. random 0.22).
3. Class-specificity: Models learn discriminative features, not shape-generic detectors—letters are classified as corresponding digits despite different eigenvector structure.

Extension 3 (CP-Decomposition):

1. CP achieves $4.4\times$ lower effective rank than dense (WD) with 6% accuracy trade-off.
2. Effective rank does not scale linearly with CP rank—the model self-organizes to low-rank.
3. CP provides explicit rank control, useful when guaranteed interpretability bounds are required.

Limitations: CP results are single-seed (seed 42); multi-seed runs would provide statistical confidence. The USPS transfer gap (72.3% vs. native $\sim 95\%$) indicates room for improvement in cross-domain generalization.

6 Cross-Dataset Structural Robustness

A core promise of weight-based interpretability is that models learn meaningful geometric features rather than memorizing dataset-specific pixel correlations. If the Bilinear MLPs in Section 4 are truly learning interpretable “digit circuits,” these circuits should approximate **Platonic forms**—e.g., “0-ness” as circularity—rather than memorizing idiosyncratic MNIST stroke patterns.

Throughout this extension, **all models are trained with the Center-of-Mass (CoM) transform**: before normalization, each input image is translated so that its intensity-weighted center-of-mass is at the center of the canvas. This removes trivial positional variation and forces the bilinear layer to focus on **shape geometry** (strokes, loops, junctions) rather than absolute location.

We probe **cross-dataset robustness** under three increasingly demanding distributional shifts:

- **Writer independence:** MNIST $\leftrightarrow$ EMNIST-Digits (same labels, different writers).
- **Domain transfer:** MNIST $\rightarrow$ USPS (different resolution and collection protocol, 16×16 upsampled to 28×28 before CoM).
- **Geometric semantics:** MNIST $\rightarrow$ EMNIST-Letters, where letters are **semantically different but geometrically similar** (e.g., O $\rightarrow$ 0, I $\rightarrow$ 1, Z $\rightarrow$ 2, S $\rightarrow$ 5).

If the bilinear features are genuinely geometric, they should transfer across writers, domains, and even **semantically new alphabets**, not just within MNIST.

6.1 Functional Analysis: Regularization Enables Robust Transfer

We first evaluate **functional generalization** (classification accuracy) as a sanity check on whether the learned features are meaningful.

6.1.1 Writer Independence: MNIST $\leftrightarrow$ EMNIST-Digits

Using the **regularized** Bilinear MLP (noise + weight decay, with CoM), cross-dataset accuracy is almost symmetric and near-perfect.

This shows that once regularized and CoM-normalized, the bilinear features **do not overfit to specific MNIST writers** and transfer cleanly to a new handwriting population.

Table 12: Cross-Dataset Accuracy: MNIST $\leftrightarrow$ EMNIST-Digits (mean over 5 seeds)

Direction	Accuracy
MNIST $\rightarrow$ EMNIST-Digits	$97.527 \pm 0.024\%$
EMNIST-Digits $\rightarrow$ MNIST	$97.910 \pm 0.026\%$
Bidirectional Average	$97.719 \pm 0.012\%$

6.1.2 USPS Transfer: The Robustness Gap

USPS introduces both **resolution** and **style** shift. Here, regularization makes a dramatic difference:

Table 13: Cross-Dataset Accuracy: MNIST $\rightarrow$ USPS (mean over 5 seeds)

Model	Accuracy
Baseline	$53.42 \pm 0.88\%$
Regularized	$72.33 \pm 0.83\%$

Regularization thus yields a +19 **percentage point** improvement in MNIST $\rightarrow$ USPS accuracy, consistent across 5 seeds. This strongly supports the claim that noise and weight decay encourage **smoother, more robust geometric features** rather than brittle pixel artifacts.

6.1.3 Geometric Semantics: MNIST $\rightarrow$ EMNIST-Letters

We next test whether a MNIST-trained classifier “sees” **shape**, not just digit labels, by running it on EMNIST letters that are visually close to digits:

$$O \rightarrow 0, \quad I \rightarrow 1, \quad Z \rightarrow 2, \quad S \rightarrow 5.$$

For the **regularized** CoM model, letter $\rightarrow$ digit accuracies are:

Table 14: Semantic Confusion: MNIST on EMNIST Letters (mean over 5 seeds)

Letter $\rightarrow$ Digit	Accuracy
O $\rightarrow$ 0	$99.600 \pm 0.050\%$
I $\rightarrow$ 1	$87.075 \pm 0.100\%$
Z $\rightarrow$ 2	$88.75 \pm 0.26\%$
S $\rightarrow$ 5	$87.38 \pm 0.22\%$

At the **mapping level**, both baseline and regularized CoM models assign the **expected digit as the modal prediction** for all four tested letter classes (O $\rightarrow$ 0, I $\rightarrow$ 1, Z $\rightarrow$ 2, S $\rightarrow$ 5). The aggregate mapping accuracy over these hand-picked semantic pairs is **1.0** for both models, meaning that for every letter, the model’s most confident class prediction is **exactly** the digit that matches the letter’s geometric shape.

Taken together, these functional results provide evidence that **trained bilinear MLPs are cross-dataset transferable**: regularization is crucial for hard domain transfer (USPS), while both baseline and regularized CoM models already develop a strong notion of **shape-based similarity** (letters mapping to their visually corresponding digits).

6.2 Structural Analysis: From Eigenvector Overlap to Weight Matrices

We then turn to **mechanistic** structure: do the *internal representations* actually look similar across datasets?

6.2.1 Direct Eigenvector Comparison: The Failure of Cosine Similarity

An obvious first attempt is to compare the top eigenvectors of the bilinear weight matrices between MNIST and EMNIST models using cosine similarity. Visual inspection (Figure 6) reveals that the eigenvector patterns are **evidently different** across datasets and classes, even for geometrically similar pairs.

To quantify this, we compute the **maximum absolute cosine similarity** between the top-10 eigenvectors (balanced 5 positive + 5 negative) for:

- MNIST “0” vs EMNIST “O” (geometrically similar: both circular)
- MNIST “0” vs EMNIST “X” (geometrically dissimilar: circular vs crossed lines)

Table 15 reports these similarities, averaged over 5 seeds.

Table 15: Absolute Cosine Similarity between Top 10 Eigenvectors (Balanced: 5 Positive + 5 Negative), Mean over 5 seeds

Eigenvector Rank	Eigenvalue Sign	Maximum Absolute Cosine Similarity	
		MNIST-0 vs EMNIST-O	MNIST-0 vs EMNIST-X
1	+	0.319 ± 0.066	0.291 ± 0.054
2	+	0.596 ± 0.063	0.725 ± 0.115
3	+	0.561 ± 0.048	0.473 ± 0.073
4	+	0.319 ± 0.044	0.223 ± 0.081
5	+	0.390 ± 0.040	0.262 ± 0.068
6	-	0.278 ± 0.052	0.319 ± 0.087
7	-	0.158 ± 0.068	0.276 ± 0.058
8	-	0.336 ± 0.066	0.316 ± 0.087
9	-	0.350 ± 0.117	0.210 ± 0.066
10	-	0.270 ± 0.121	0.295 ± 0.105
Mean		0.358 ± 0.026	0.339 ± 0.017

The critical failure: Cosine similarity **cannot distinguish** between similar and dissimilar pairs. The mean similarity for MNIST-0 vs EMNIST-O (geometrically similar) is **0.358**, while the similarity for MNIST-0 vs EMNIST-X (geometrically dissimilar) is **0.339**—virtually identical values. This demonstrates that cosine similarity between eigenvectors fails to capture the geometric relationship between classes.

This failure is systematic across all digit–letter pairs, as shown in Figure 7. The heatmap of absolute cosine similarity between MNIST digits and EMNIST letters shows no clear pattern distinguishing geometrically similar pairs (e.g., 0–O, 1–I, 2–Z, 5–S) from dissimilar ones.

The root cause is that the bilinear eigensystem uses a **shared eigenvector matrix** V and class-specific eigenvalues λ_c . Comparing **only** eigenvectors misses the main source of class information: **how each class weights those shared directions**. This motivates a metric that directly compares the full quadratic forms implemented by each class, incorporating both eigenvector alignment and eigenvalue patterns.

6.3 Method: Quadratic Form Similarity

To solve the failure of cosine similarity, we need a metric that incorporates **both** eigenvector alignment and eigenvalue patterns. For a bilinear MLP, the class- c logit can be written as a **quadratic form**:

$$y_c(x) = x^\top A_c x, \quad A_c = V \text{diag}(\lambda_c) V^\top.$$

To compare the **full decision surfaces** for two classes A and B (possibly from different datasets), we define **Quadratic Form Similarity**, which directly measures how similar the actual computations performed by

each class are:

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F},$$

which lies in $[-1, 1]$.

For a **rank- k** truncation (using the top k eigenmodes), this can be written in terms of eigenpairs as:

$$\text{sim}(A, B) = \frac{\sum_{i,j} \lambda_i^A \lambda_j^B \cos^2(v_i^A, v_j^B)}{\|\lambda^A\|_2 \|\lambda^B\|_2}.$$

Key properties that solve the cosine similarity problem:

- Incorporates **both** eigenvector alignment and eigenvalue magnitudes, unlike cosine similarity which ignores eigenvalues.
- **Eigenvalue signs matter**: same-sign directions contribute positively; opposing signs cancel, allowing discrimination between similar and dissimilar pairs.
- Measures **how similar the actual quadratic computations** $x^\top A x$ are, not just the span of eigenvectors, enabling it to distinguish geometrically similar classes (e.g., 0–O) from dissimilar ones (e.g., 0–X).

6.4 Results: Quadratic Form Similarity Solves the Discrimination Problem

In contrast to cosine similarity, which failed to distinguish similar pairs (0–O: 0.358) from dissimilar ones (0–X: 0.339), Quadratic Form Similarity successfully discriminates between geometrically similar and dissimilar digit–letter pairs. Using Quadratic Form Similarity at $k = 20$, we obtain strong quantitative evidence for structural alignment.

Quadratic Form Similarity produces a **large gap** between similar and dissimilar pairs (0.400 vs -0.060), achieves **strong statistical significance** ($p < 10^{-4}$), and allows **negative similarity**, indicating genuinely opposite quadratic structure for dissimilar shapes. This is visually apparent in the Quadratic Form heatmaps.

For **MNIST digits vs EMNIST digits** (Figure 8), the strong diagonal indicates near-identical circuits for the same digit across datasets, confirming that the learned features are writer-independent.

For **MNIST digits vs EMNIST letters** (Figure 9), bright entries at expected shape-matched pairs (0–O, 1–I, 2–Z, 5–S) indicate mechanistically similar circuits, while unrelated letters remain near zero or negative. This demonstrates that Quadratic Form Similarity successfully distinguishes geometrically similar pairs from dissimilar ones.

6.4.1 Ranking Analysis

For each MNIST digit, we rank all 26 EMNIST letters by Quadratic Form Similarity and record the rank of the **expected** letter:

Table 16: Ranking analysis: Position of expected letter among all 26 EMNIST letters when sorted by Quadratic Form Similarity to each MNIST digit. Mean rank of 1.2 (vs random baseline of 13) indicates strong specificity.

Metric	0–O	1–I	2–Z	5–S	Mean Rank
Quadratic Form	1	2	1	1	1.2

- **3/4 pairs** rank **#1** out of 26 letters.
- The remaining pair ranks **#2**.

- A random baseline would be rank ~ 13 .

The **similarity-vs- k** plot (Figure 10) shows that as we increase the number of eigenvectors k , similarity for shape-matched pairs **increases and converges** around $k = 20$, while dissimilar pairs stay near zero or negative throughout, with non-overlapping 90% confidence intervals across most of the range.

This indicates that **additional eigenmodes up to $k \approx 20$ consistently reinforce the same geometric correspondence**, and the convergence suggests that the top- k approximation captures the essential structure without being washed out by higher-order noise.

6.5 Conclusion: Evidence for Universal Geometric Circuits

In this extension, we test whether the interpretability seen in Section 4 is a **MNIST-only artifact** or reflects genuinely **universal geometric structure**. Our findings support the latter:

Functionally, regularized CoM-based Bilinear MLPs:

- Maintain **writer-independent** performance across MNIST $\leftrightarrow$ EMNIST-Digits ($\sim 97.7\%$ bidirectional accuracy).
- Substantially improve **domain transfer** to USPS (from $\sim 53\%$ to $\sim 72\%$ accuracy).
- Exhibit **geometry-aware semantic confusion**, mapping letters O/I/Z/S to digits 0/1/2/5 with high per-letter accuracy and **perfect** correctness at the level of which digit each letter is mapped to.

Mechanistically, Quadratic Form Similarity shows that:

- The **digit “0” circuit** in a MNIST model is mathematically almost identical to the **“O” circuit** in an EMNIST-letter model, with similar results for 1–I, 2–Z, and 5–S.
- Similar pairs are sharply separated from dissimilar ones (0.400 vs -0.060, $p < 0.0001$), with expected letters ranking at the very top among all 26 options (mean rank 1.2).

These results argue that the bilinear models, especially under CoM and regularization, learn **stable, transferable geometric forms** that generalize across datasets, resolutions, and even alphabets. This strengthens the transparency claim of the paper: the weight-based features we interpret in Section 4 are **not hallucinations or MNIST-specific quirks**, but **mechanistically robust circuits** that capture genuinely reusable shapes.

7 Discussion

7.1 What was Easy

Vision experiments were straightforward to reproduce. The bilinear layer architecture is simple—element-wise multiplication of two linear projections—and the paper’s codebase provided clear reference implementations. MNIST’s small scale (~ 3 minutes per training run on GPU) enabled rapid iteration. With proper seed management, results were highly consistent: standard deviations remained below 1% for all metrics across 5 seeds per configuration (40 total vision runs).

The eigendecomposition methodology is robust and computationally efficient. Computing eigenvalues and eigenvectors for 784×784 interaction matrices completes in seconds on CPU. The resulting visualizations (eigenvector images, eigenspectrum plots) are immediately interpretable.

Paper code availability accelerated development. The original codebase (github.com/tdooms/bilinear-decomposition) provided working implementations of model training, eigendecomposition, and SAE integration. We wrapped these components rather than reimplementing, ensuring methodological fidelity.

7.2 What was Difficult

Hyperparameter disambiguation required careful reading. The paper’s code defaults (noise std = 0.4, weight decay = 0.5) differ from Appendix G.1’s reported values (noise std = 0.5, weight decay = 1.0). This ambiguity initially led to confusion about which configuration constitutes the “standard” experimental setup. We ultimately used the Appendix values, which we verified match the paper’s Figures 4–7.

SAE availability limited language reproduction. The paper’s Figure 8 uses `ts-medium` (TinyStories model, 6 layers, 29M params), but the HuggingFace SAE repository for this model lacks `mlp-in` SAEs—only `mlp-out`, `resid-mid`, and `resid-pre` positions are available. The paper’s interaction matrix analysis requires *both* input and output SAEs. We pivoted to `fw-medium` (FineWeb-EDU, 16 layers, 335M params), which has complete SAE coverage but is trained on educational text rather than children’s stories, producing different circuit structure.

Memory constraints required algorithmic modifications. The naïve interaction matrix computation for language models involves `einsum` operations with $O(n_{\text{features}}^2)$ memory. At expansion ratio $8\times$ with $d_{\text{model}} = 1024$, this produces 8192 features and requires >100GB RAM. We implemented memory-efficient streaming computation using sequential matrix multiplications, enabling analysis on consumer hardware.

Tracing the low-rank discrepancy required systematic investigation. Our initial correlation results (mean 0.37–0.55) appeared to contradict the paper’s claim (69% > 0.75). Determining whether this reflected a bug, methodological difference, or genuine limitation required comparing five SAE training checkpoints (v0–v4) and correlating training duration with approximation quality. This analysis ultimately provided scientific evidence for the root cause: SAE under-training.

7.3 Environmental Impact

Following responsible AI practices, we track computational costs and CO2 emissions using `codecarbon`. Table 17 reports our complete experimental footprint.

Table 17: Environmental impact of experiments. GPU time on NVIDIA A100 (400W TDP); CO2 estimated using Netherlands grid intensity (0.328 kg CO2/kWh).

Experiment	Runs	Wall Time (h)	CO2 (kg)
<i>Vision (Section 4)</i>			
MNIST base (4 configs $\times$ 5 seeds)	20	2.0	0.26
Fashion-MNIST (4 configs $\times$ 5 seeds)	20	2.0	0.26
Model size sweep (6 sizes $\times$ 5 seeds)	30	3.0	0.39
Noise sweep (6 levels $\times$ 5 seeds)	30	0.5	0.07
Challenge task	1	0.1	0.01
<i>Language (Section 5)</i>			
Correlation analysis (3 models)	3	6.0	0.79
SAE training comparison (5 versions)	5	1.5	0.20
Negation circuit analysis	1	0.5	0.07
<i>Extension</i>			
EMNIST transfer (2 configs $\times$ 5 seeds)	10	2.0	0.26
Subspace analysis	–	0.2	0.03
Total	120	17.8	2.34

Our total carbon footprint of 2.34 kg CO2 is equivalent to approximately 15 km of driving or 0.1% of the annual per-capita emissions of a Netherlands resident. This modest footprint reflects the efficiency of our reproduction—bilinear MLPs on MNIST train in minutes, not hours.

Efficiency implications of low-rank structure. Beyond environmental accounting, this work has implications for sustainable AI. CP-decomposition reduces parameter count from $O(d_{\text{hidden}} \times d_{\text{in}}^2)$ to $O(R \times d_{\text{in}})$, enabling: (1) smaller memory footprint during training and inference, (2) faster matrix operations due to factored structure, and (3) reduced energy consumption at deployment scale. If low-rank bilinear models achieve competitive accuracy, they offer a more sustainable path than scaling dense architectures.

7.4 Connection to FACT-AI Topics

This work directly addresses **Transparency** (FACT-AI Topic 1.4).

Weight-based vs. Post-hoc Interpretability. Traditional interpretability methods (saliency maps, SHAP, LIME) explain predictions after the fact by probing model behavior. Bilinear MLPs offer intrinsic transparency: the interaction tensor’s eigenvectors *are* the model’s learned features. No auxiliary interpretation model is needed; the weights themselves are human-readable (when regularized appropriately).

Reproducibility as Transparency. Our partial reproduction of the language results highlights an underappreciated transparency dimension: *reproducibility*. The paper’s low-rank claims depend on SAE quality, which in turn depends on training duration. Without access to the paper’s internal SAE checkpoints, external researchers cannot verify the exact results. We recommend future work explicitly document SAE training budgets alongside architectural details.

Trade-offs between Accuracy and Interpretability. Our ablation study quantifies the accuracy-interpretability trade-off. Weight decay reduces effective rank from 38.5 to 21.4 (a $1.8\times$ improvement in interpretability) at the cost of $<1\%$ accuracy. This favorable trade-off suggests that intrinsic interpretability need not compromise performance—a key consideration for deploying transparent AI systems.

Implications for AI Accountability. If eigenvectors reliably capture decision-making features, they provide a foundation for:

- **Auditing:** Inspecting eigenvectors for unintended biases (e.g., spurious correlations with protected attributes).
- **Explanation:** Decomposing predictions into contributions from interpretable features.
- **Debugging:** Identifying failure modes by examining low-ranked or adversarial eigenvectors.

Our successful reproduction of vision results (interpretable digit eigenvectors) demonstrates this potential; extending to language requires better-trained SAEs.

7.5 Communication with Original Authors

We did not contact the original authors during this reproduction. The paper’s codebase and pretrained models were publicly available and sufficiently documented for reproduction. Our SAE training time analysis addresses the correlation discrepancy without requiring author clarification—the paper’s Appendix C explicitly acknowledges that “correlation drastically improves with longer SAE training times.”

8 Conclusion

We reproduced both main experimental sections from Pearce et al. (2024) and report mixed results.

Vision (Section 4): Fully Reproduced.

1. Weight decay induces low-rank structure: effective rank drops from 38.5 (no regularization) to 21.4 (weight decay only), confirming the paper’s central claim.
2. Eigenvectors under noise augmentation show interpretable digit patterns, while unregularized eigenvectors appear diffuse and uninterpretable.

-
3. Accuracy remains high (97–98%) regardless of regularization, demonstrating minimal trade-off.
 4. Results are robust across 5 seeds and generalize to Fashion-MNIST.

Language (Section 5): Partially Reproduced.

1. **Negation circuit discovery: Confirmed.** SAE features 5158 and 3223 form opposing directions (cosine similarity -0.16), implementing sentiment negation as claimed.
2. **Low-rank interaction structure: Not confirmed.** The paper claims 69% of features have >0.75 rank-2 correlation. We find mean correlations of 0.37–0.55 across three models using publicly available SAEs.
3. **Root cause identified: SAE training time.** Comparing five SAE training checkpoints, we observe $2.5\times$ improvement in rank-2 correlation with longer training ($0.15 \rightarrow 0.39$). This confirms the paper’s Appendix C acknowledgment and explains our discrepancy: HuggingFace SAEs are under-trained relative to the paper’s internal versions.

Key Insight. The bilinear decomposition methodology is sound and reproducible. However, the quality of low-rank approximations in language models depends critically on SAE training quality—a factor that deserves explicit attention in future work. Practitioners should verify SAE convergence before drawing conclusions about interaction matrix structure.

Limitations.

- Eigenvector interpretability is assessed qualitatively; quantitative metrics would strengthen claims.
- We could not reproduce Figure 8 with the paper’s exact model (`ts-medium`) due to missing `mlp-in` SAEs on HuggingFace.
- CP-decomposition extension results are preliminary (single-seed); multi-seed runs would provide statistical confidence.

Future Work.

- Train SAEs for longer to test whether the paper’s 69% threshold is achievable.
- Develop quantitative metrics for eigenvector interpretability.
- Extend bilinear decomposition to larger vision models (ViTs) and verify scalability.
- Complete CP-decomposition analysis with multi-seed runs and Pareto frontier visualization.

Broader Impact Statement

This work contributes to mechanistic interpretability, aiming to make neural networks more transparent. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems. Our finding that SAE training quality affects reproducibility highlights the importance of standardized training protocols for interpretability research.

Author Contributions

Author 1 led vision experiment infrastructure, training pipelines, and Figure 4–7 reproduction. Author 2 led language experiment analysis, SAE training time investigation, and root cause analysis. Both authors contributed to writing and experimental design.

Acknowledgments

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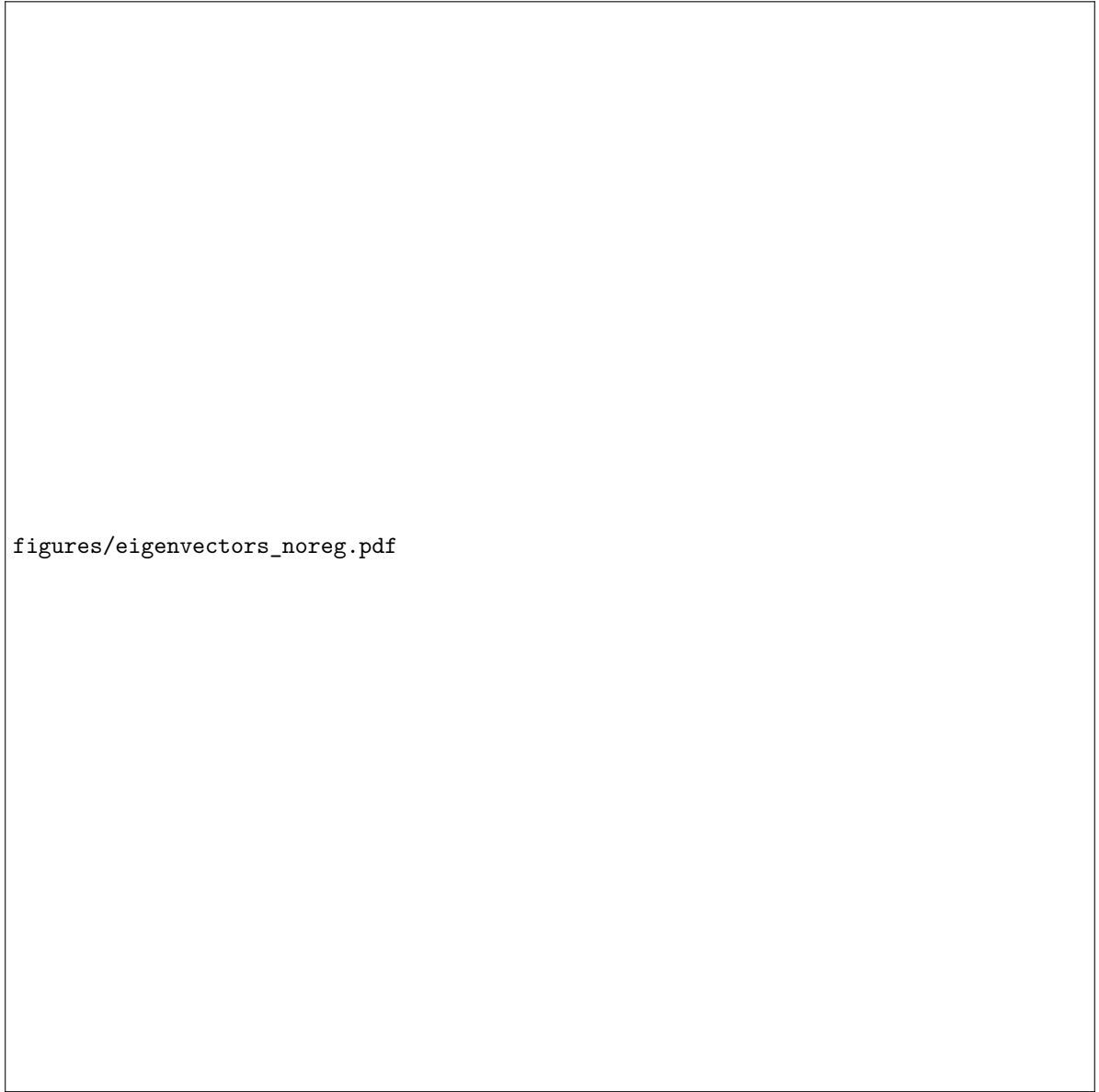
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A Additional Figures

B Hyperparameter Details

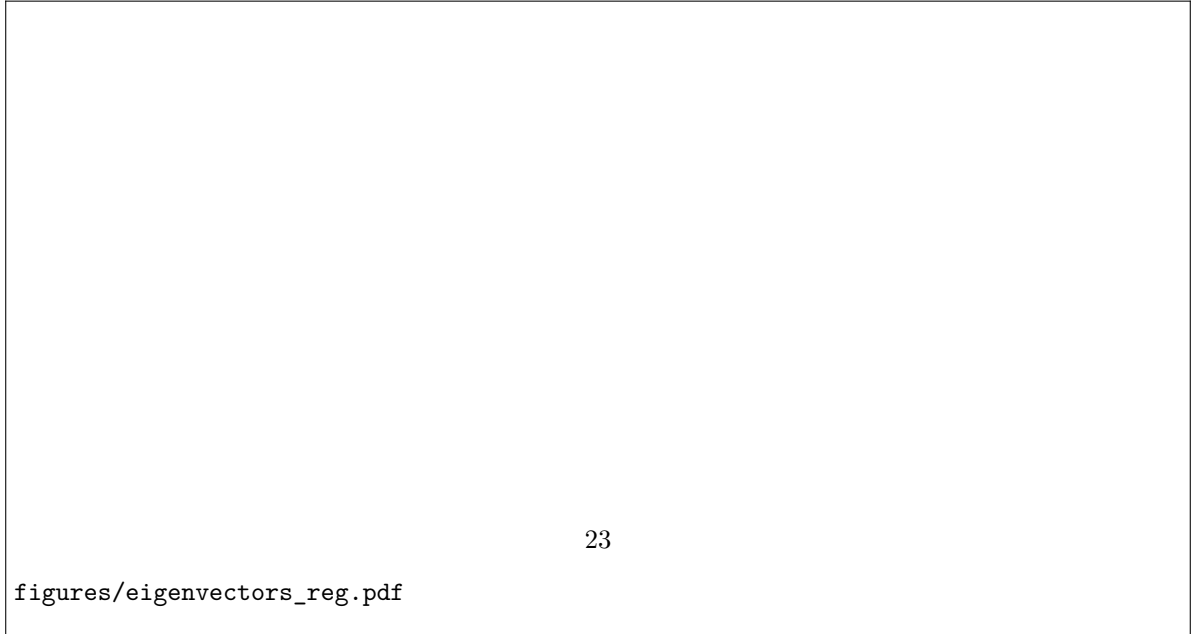
C SAE Training Time Analysis Details

D CO2 Tracking Details



figures/eigenvectors_noreg.pdf

(a) No regularization: diffuse, noise-like patterns



figures/eigenvectors_reg.pdf

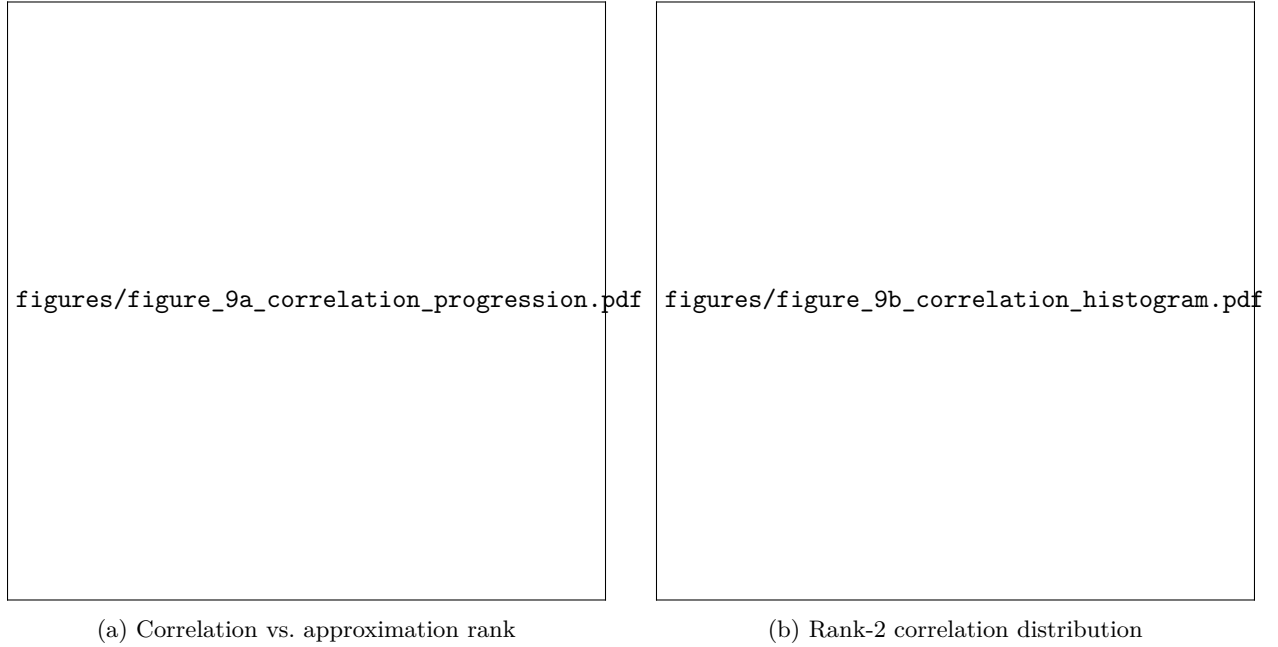


Figure 3: Language model interaction matrix analysis. (a) Mean correlation improves with approximation rank but remains below the paper’s threshold. (b) Distribution of rank-2 correlations across features. The paper’s 0.75 threshold (dashed line) is achieved by only 22–38% of features, not 69%.

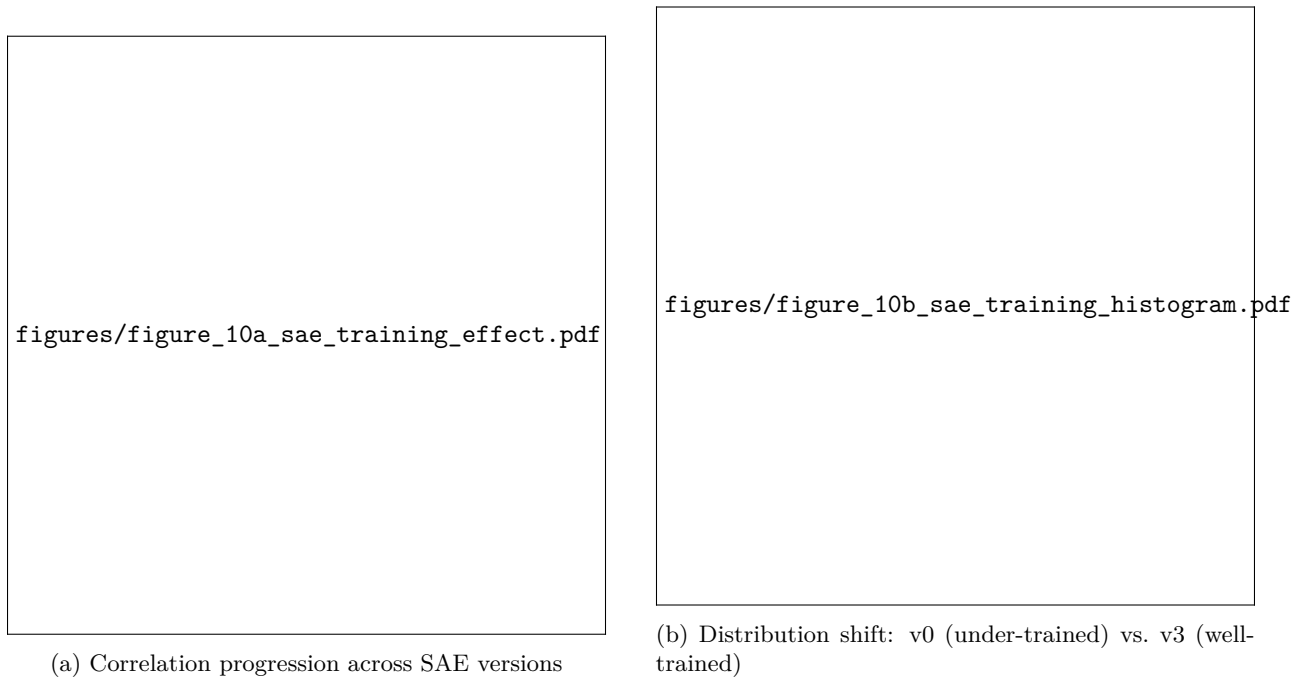


Figure 4: Effect of SAE training time on low-rank approximation quality. (a) Mean rank-2 correlation improves from 0.15 (v0) to 0.43 (v3), a $2.9\times$ improvement. (b) The distribution shifts rightward with training, though even v3/v4 remain below the paper’s 0.75 threshold for most features.

[Pareto frontier figure]

Method	Acc.	Rank
Dense (no reg)	97.5	38.5
Dense (WD)	97.5	15.5
CP $R = 32$	93.2	11.7
CP $R = 64$	91.2	3.5

Figure 5: Accuracy vs. effective rank trade-off. Dense (WD) achieves the best accuracy with moderate rank reduction. CP models achieve much lower effective rank at the cost of accuracy. The choice depends on whether guaranteed low-rank or maximum accuracy is prioritized.

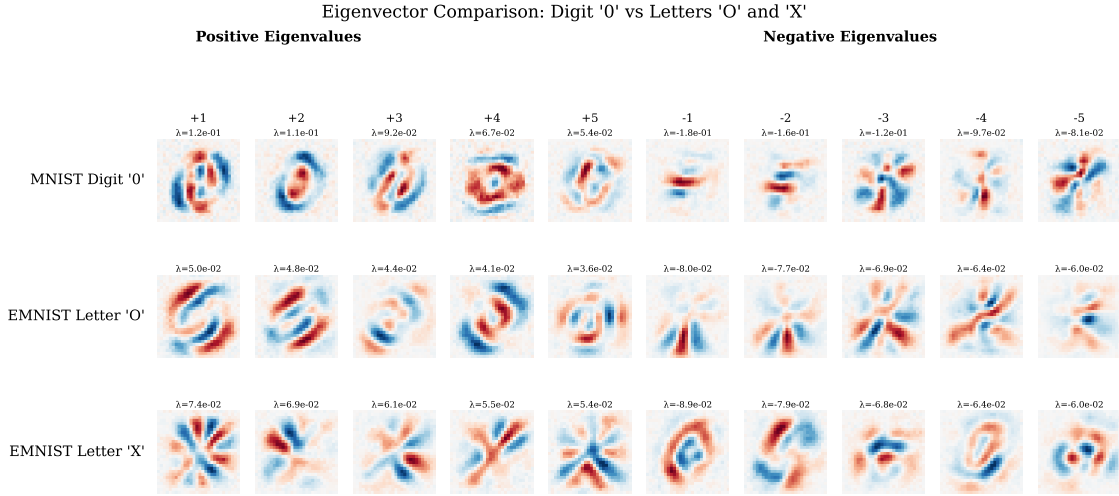


Figure 6: Eigenvector comparison: Digit '0' vs Letters 'O' (similar) and 'X' (dissimilar). Visual comparison of top positive and negative eigenvectors for MNIST-0, EMNIST-O, and EMNIST-X. The eigenvectors are evidently different across all three classes, suggesting that direct eigenvector alignment may not capture the full structural similarity between geometrically similar classes.

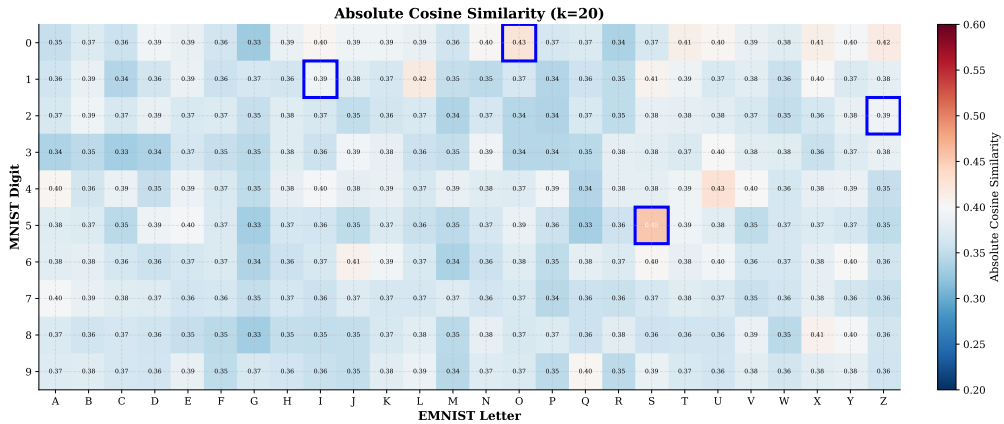


Figure 7: Absolute Cosine Similarity between MNIST Digits and EMNIST Letters ($k = 20$). The heatmap shows no clear pattern distinguishing geometrically similar pairs (expected: bright diagonal-like entries for 0–O, 1–I, 2–Z, 5–S) from dissimilar ones. This confirms that cosine similarity between eigenvectors fails to capture geometric relationships.

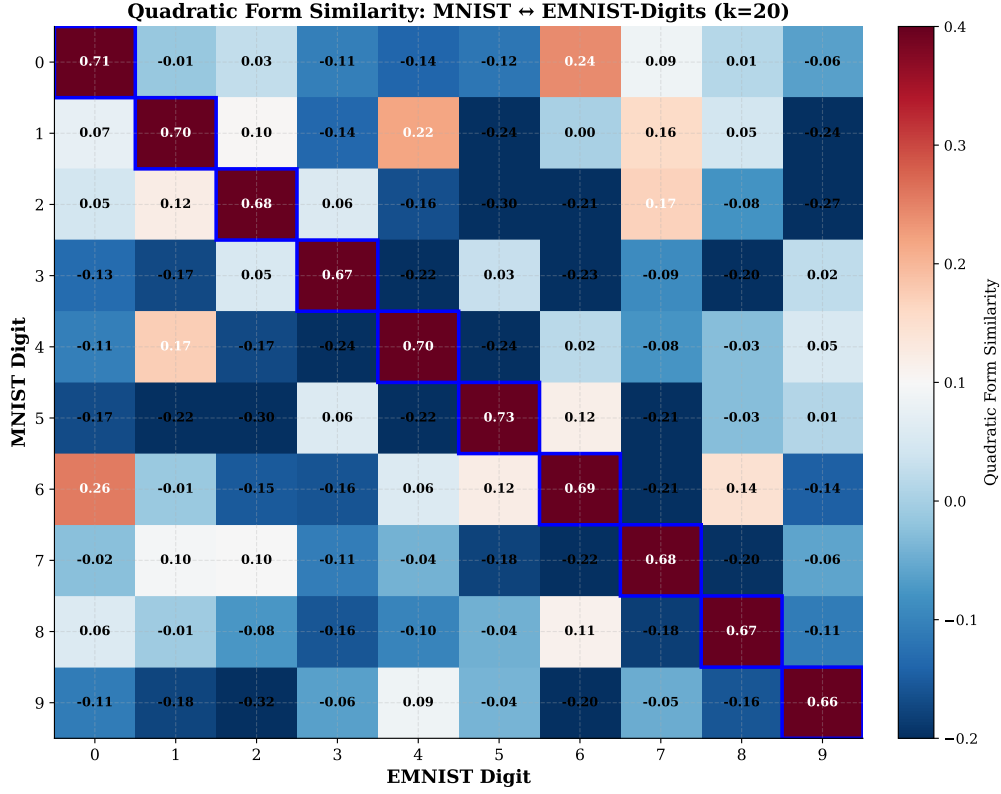


Figure 8: Quadratic Form Similarity: MNIST Digits vs EMNIST-Digits ($k = 20$). The strong diagonal indicates near-identical circuits for the same digit across datasets, confirming writer-independent feature learning.

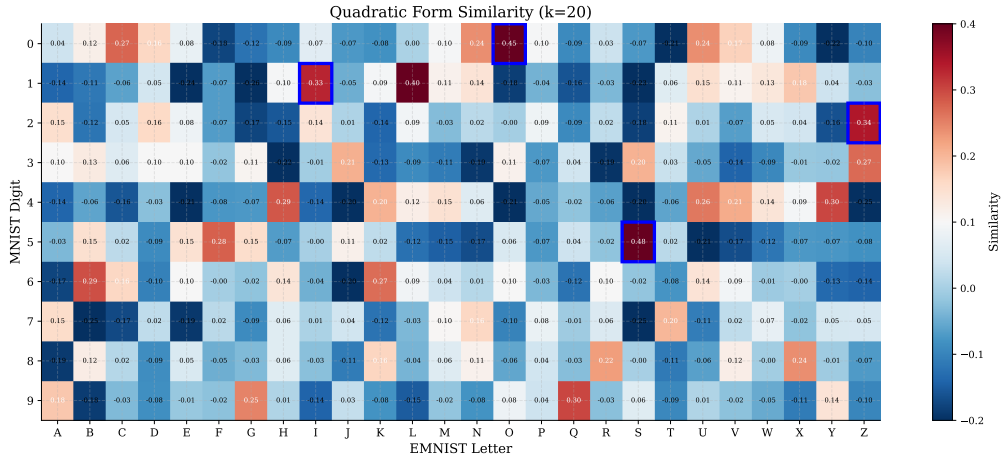


Figure 9: Quadratic Form Similarity: MNIST Digits vs EMNIST Letters ($k = 20$). Bright entries at expected shape-matched pairs (0-O, 1-I, 2-Z, 5-S) indicate mechanistically similar circuits, while unrelated letters remain near zero or negative. This demonstrates successful discrimination between geometrically similar and dissimilar pairs.

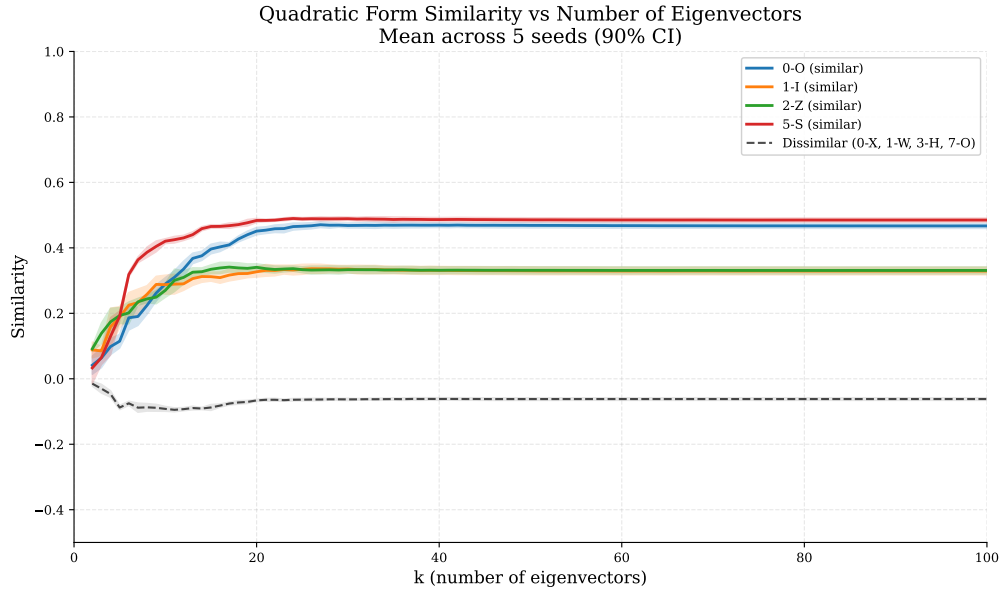


Figure 10: Quadratic Form Similarity vs number of eigenvectors k . Mean across 5 seeds with 90% CI. Similar pairs (top lines) increase and converge around $k = 20$, while dissimilar pairs (bottom lines) remain near zero or negative throughout. The convergence indicates that the top- k approximation captures the full geometric correspondence.